CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 41 (2013) 150–155

Contents lists available at SciVerse ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic assessment of the B–Ce and B–Pr systems

C.P. Wang [a], Y. Shi [a], D. Wang [a], Y. Lu [a], D.L. Zhao [b], X.J. Liu [a,*]

[a] Department of Materials Science and Engineering, College of Materials, and Research Center of Materials Design and Application, Xiamen University, Xiamen 361005, PR China

[b] Research Institute of Functional Materials, Central Iron & Steel Research Institute, Beijing 100081, PR China

ARTICLE INFO

Article history:
Received 18 January 2013
Received in revised form
26 March 2013
Accepted 27 March 2013
Available online 17 April 2013

Keywords:
B–Ce system
B–Pr system
Phase diagrams
Thermodynamic modeling

ABSTRACT

By using the Calculation of Phase Diagrams (CALPHAD) technique, the thermodynamic assessments of the B–Re (Re: Ce, Pr) binary system were carried out based on the experimental data including thermodynamic properties and phase equilibria. Gibbs free energies of the solution phases (liquid, fcc, bcc, dhcp) were modeled by the subregular solution model with the Redlich–Kister equation, and those of the intermediate compounds (B_4Ce, B_6Ce, B_5Pr_2, B_4Pr, B_6Pr) were described by the two-sublattice model. A consistent set of thermodynamic parameters leading to reasonable agreement between the present calculated results and experimental data was obtained.

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1. Introduction

The rare earth elements exhibiting good combination of magnetic, optical, electrical and thermal properties have been paid increasing attention and widely used to prepare new high-performance functional materials, such as permanent magnet, magnetic refrigerant and hydrogen storage material [1–3]. The borides of rare earth metals (REM) constitute a group of materials which possess unique and outstanding chemical and physical properties [4,5]. The CALPHAD method, which is a powerful tool to reduce time and cost during development of materials [6], can effectively provides a clear guideline for materials design. In order to develop the thermodynamic database of phase equilibria in the B-REM multicomponent alloy systems, the thermodynamic description of each lower-order system which forms a part of this thermodynamic database is necessary. The purpose of the present work is to present the thermodynamic descriptions of the B–Re (Re: Ce, Pr) binary system by the CALPHAD method. The thermodynamic parameters of each phase in the B–Re (Re: Ce, Pr) system are optimized by fitting the experimental data on the thermodynamic properties and phase equilibria.

2. Thermodynamic models

The information of the stable solid phases and the used models in the B–Re (Re: Ce, Pr) system is listed in Table 1.

2.1. Solution phases

In the B–Ce and B–Pr systems (B–Re system), the Gibbs free energies of the solution phases including liquid, fcc, bcc, hcp and dhcp were described by the subregular solution model with the Redlich–Kister formula [7]. The molar Gibbs free energy of each solution phase in B–Re system can be expressed as

$$G_m^\phi = \sum_{i=B,\mathrm{Re}} {}^0G_i^\phi x_i + RT \sum_{i=B,\mathrm{Re}} x_i \ln x_i + x_B x_{\mathrm{Re}} \sum_{m=0}^{n} {}^mL_{B,\mathrm{Re}}^\phi (x_B - x_{\mathrm{Re}})^m + \Delta^{mag} G_m^\phi, \quad (1)$$

where ${}^0G_i^\phi$ is the molar Gibbs free energy of pure component i in the respective reference state with the structure ϕ in a nonmagnetic state, which is taken from the compilation by Dinsdale [8]. x_i Denotes the mole fraction of the component i; R is the gas constant; T is the absolute temperature; and ${}^mL_{B,\mathrm{Re}}^\phi$ is the interaction energy between B and Re atoms, and is expressed as

$${}^mL_{B,\mathrm{Re}}^\phi = a + bT, \quad (2)$$

where the parameters of a and b are optimized based on the experimental data in the present work.

In the B–Ce and B–Pr systems, the magnetic contributions to the Gibbs free energies are considered in the bcc, fcc and dhcp phases. $\Delta^{mag}G_m^\phi$ Is the magnetic contribution to the Gibbs free energy, which is described by the following equation [9]:

$$\Delta^{mag} G_m^\phi = RT \ln(\beta + 1) f(\tau), \quad (3)$$

where τ is defined as T/T_C^ϕ, T_C^ϕ is the Curie temperature of solution for ferromagnetic ordering and β is the Bohr magneton number

* Corresponding author. Tel.: +86 592 2187888; fax: +86 592 2187966.
E-mail address: lxj@xmu.edu.cn (X.J. Liu).

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http://dx.doi.org/10.1016/j.calphad.2013.03.001

Table 1
Information on the stable solid phases and the used models in the B–Ce and B–Pr systems [4,5].

System	Phase	Prototype	Structure report designation	Modeling phase	Used models
B–Ce	(δCe)	Cu	A2	(B, Ce)$_1$	SSM
	(γCe)	W	A1	(B, Ce)$_1$	SSM
	B_4Ce	B_4Th	$D1_e$	(B)$_4$(Ce)	SM
	B_6Ce	CaB_6	$D2_l$	(B)$_6$(Ce)	SM
	(βB)	βB	–	(B, Ce)$_1$	SSM
B–Pr	(βPr)	W	A2	(B, Pr)$_1$	SSM
	(αPr)	La	A3'	(B, Pr)$_1$	SSM
	B_5Pr_2	–	–	(B)$_5$(Pr)$_2$	SM
	B_4Pr	B_4Th	$D1_e$	(B)$_4$(Pr)	SM
	B_6Pr	CaB_6	$D2_l$	(B)$_6$(B, Pr)	TSM
	(βB)	βB	–	(B, Pr)$_1$	SSM

SSM: subregular solution model.
SM: sublattice model.
TSM: two-sublattice model.

per mole. The function $f(\tau)$ is formulated by the polynomial of the normalized temperature, as follows:

$$f(\tau)=1-\frac{1}{D}\left[\frac{79\tau^{-1}}{140p}+\frac{474}{497}\left(\frac{1}{p}-1\right)\left(\frac{\tau^3}{6}+\frac{\tau^9}{135}+\frac{\tau^{15}}{600}\right)\right],\quad for\le 1 \tag{4}$$

$$f(\tau)=-\frac{1}{D}\left(\frac{\tau^{-5}}{10}+\frac{\tau^{-15}}{315}+\frac{\tau^{-25}}{1500}\right),\quad for>1 \tag{5}$$

where

$$D=\frac{518}{1125}+\frac{11692}{15975}\left(\frac{1}{p}-1\right),$$

p depends on the structure, 0.40 for bcc structure and 0.28 for others.

2.2. Stoichiometric intermediate compounds

The intermediate compounds, B_4Ce, B_6Ce, B_5Pr_2, and B_4Pr in the B–Re (Re: Ce, Pr) binary system are treated as stoichiometric phases, as no experimental data was made available for the homogeneity range of each compound. The Gibbs free energy per mole of formula unit $(B)_m(Re)_n$ can be expressed as the following equation referring to the pure element in their nonmagnetic state

$$G^{B_mRe_n}_{B:Re}=\Delta^0G^{B_mRe_n}_{B:Re}+m^0G^{ref}_B+n^0G^{ref}_{Re}+\Delta^{mg}G^{B_mRe_n}_m, \tag{6}$$

where the $\Delta^0G^{B_mRe_n}_{B:Re}$ represents the standard Gibbs free energy of formation per mole of formula unit B_mRe_n referred to the standard element reference (SER) state of the component elements, which is described as

$$\Delta^0G^{B_mRe_n}_{B:Re}=a'+b'T, \tag{7}$$

where the parameters of a' and b' are evaluated in the present work.

In the B–Ce and B–Pr systems, the magnetic contributions to the Gibbs free energies for the B_4Ce, B_6Ce, B_5Pr_2, and B_4Pr compounds are considered. $\Delta^{mg}G^{B_mRe_n}_m$ is the magnetic contribution to the Gibbs free energy, which is similarly expressed by Eqs. (3)–(5), where the values of β and T^{ϕ}_C of each intermediate compound are taken from the literature [1].

2.3. Non-stoichiometric intermediate phase

The B_6Pr phase in the B–Pr binary system was treated as a non-stoichiometric intermediate compound. The Gibbs free energy of the B_6Pr phase was described by the two-sublattice model [10]. The Gibbs free energy of formation per mole of formula unit $(B)_6(B, Pr)$ can be expressed as the following equation with reference to the pure elements in their nonmagnetic state

$$G^{B_6Pr}=y^{II}_B{}^0G^{B_6Pr}_{B:B}+y^{II}_{Pr}{}^0G^{B_6Pr}_{B:Pr}+0.1429RT(y^{II}_B\ln y^{II}_B+y^{II}_{Pr}\ln y^{II}_{Pr})$$
$$+0.1429y^{II}_By^{II}_{Pr}L_{B:B,Pr}, \tag{8}$$

where y^{II}_B and y^{II}_{Pr} are the site fractions of component B and Pr located on the second sublattice. The parameters ${}^0G^{B_6Pr}_{B:*}$ represent the Gibbs free energy of the compound phase when the first sublattice is occupied by element B, while the second sublattice is completely occupied by element B or Pr, respectively. $L_{B:B,Pr}$ is the interaction parameter between B and Pr atoms in the second sublattice, when the first sublattice is occupied by element B. ${}^0G^{B_6Pr}_{B:B}$, ${}^0G^{B_6Pr}_{B:Pr}$ and $L_{B:B,Pr}$ were evaluated in the present work.

3. Experimental information

3.1. The B–Ce system

The phase diagram of the B–Ce system reviewed by Liao et al. [11] is shown in Fig. 1. The B–Ce system is consisted of three solid solution phases (δCe (bcc), γCe (fcc) and βB (rhombohedral)) and two REM borides (B_4Ce and B_6Ce phases). Since the solubility of B in fcc (δCe) or bcc (γCe) and that of Ce in rhombohedral (βB) were not reported in Ref. [11], and no other literatures have reported the measurement of solubility of terminal solid solutions, both (βB) and (Ce) solid solutions have negligible composition ranges. Stecher et al. [12] reported that B_4Ce decomposed peritectically, as hypothesized by Brewer et al. [13], and gave the decomposition temperature (2375 °C), the melting point of B_6Ce (2540 °C) and the (B_6Ce)–(βB) eutectic temperature (2200 °C). Stecher et al. [12] reported the partial B–Ce phase diagram, and several authors [13–18] also modified this diagram. The melting point of the pure boron in the experimental phase diagram has been reported as 2042 °C by Mar [14], 2074 °C by Kimpel and Moss [15], and 2092 °C by Holcombe et al. [16]. However, Stecher showed the melting temperature for B in the experimental phase diagram (~2300 °C) is 208 °C higher than the melting point reported by Spear [4,5], making the other temperature measurements suspect as well. In the present work, we used 2075 °C as the melting temperature of the pure boron, which was compiled in the SGTE-Pure database [8]. For the melting temperature of rare-earth boride B_4Ce, Stecher et al. showed a value as 2380 °C, Smith [17] determined lower

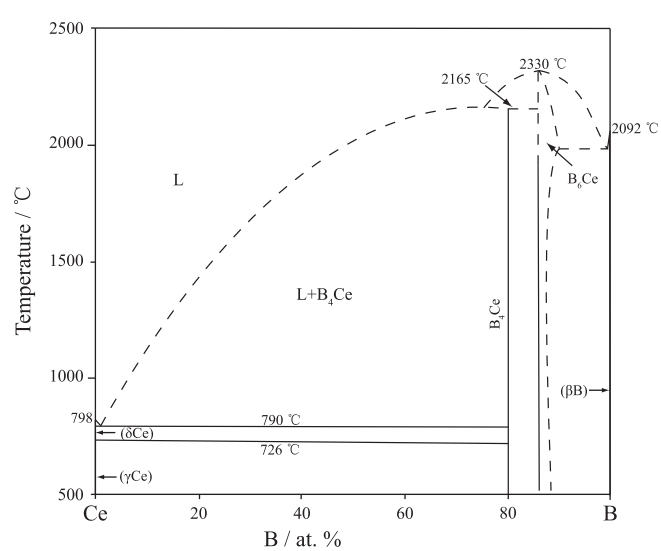

Fig. 1. The phase diagram of the B–Ce system reviewed by Liao et al. [11].

limits of 2100 °C. For the melting temperature of rare-earth boride B_6Ce, von Stackelberg and Neumann [18] reported a value of 2190 °C, this value is unreasonable low in comparison with the 2550 °C value given by Stecher et al. [12].

Liao et al. [11] recently remodified the phase diagram of B–Ce system, which is characterized by stable boride compounds, negligible solid solubility ranges and there is minimal experimental data for liquidus temperatures and compositions [5]. The congruently melting temperature of compound B_6Ce is 2330 °C. The temperature of the peritectic reaction of the L+B_6Ce↔B_4Ce is 2165 °C. There is a peritectic reaction of the L+(δCe)↔(γCe) at 726 °C, and a eutectic reaction of the L↔(δCe)+B_4Ce at 790 °C in the B–Ce system. The temperature of the L↔B_6Ce+(βB) reaction was given to be 1990 °C, which largely differs from what reported by Stecher et al. [12].

In addition, the standard enthalpy of formation of the intermediate compound of the B_6Ce was measured by Meschel and Kleppa [19] using direct synthesis calorimetry (DSC).

3.2. The B–Pr system

The phase diagram of the B–Pr system reviewed by Moffatt [20] is shown in Fig. 2. The phase diagram of the B–Pr system was originally constructed by Spear [4,5] and Moffatt [20]. The B–Pr system is consisted of three terminal solid solution phases (βPr (bcc), αPr (dhcp) and βB (rhombohedral)) and three intermediate compounds (B_5Pr_2, B_4Pr and B_6Pr phases). Since no literatures have reported the measurement of solubility of terminal solid solutions, the solubility of each element in the B–Pr terminal solid solutions was not taken into account in the present optimization. Spear [4,5] and Moffatt [20] reported that the melting temperature of B_4Pr is close to 2350 °C, while Smith and Gilles [17] determined the lower limits of the melting temperature which is 2250 °C, and also reported 2250 °C for the melting temperature of B_6Pr. Mordovin and Timofeeva [21] measured and predicted the temperature of B_4Pr as 2350 °C and 2610 °C for B_6Pr. By using the single-crystal X-ray diffraction analysis method, Mikhalenko and Kuz' ma [22] determined the crystal structure of the lower boride of praseodymium, whose composition is described by B_5Pr_2 [23]. Spear did not determine the exact value for the melting temperature of B_5Pr_2 and its melting temperature was estimated to be 2000 ± 300 °C [4,5]. The liquidus curves were estimated by correlating the melting behavior of the B–Pr system with the other REM boride systems [4,5]. The B–Pr phase diagram shows one congruently melting phase B_6Pr and has a narrow range of homogeneity, and several peritectic melting phases, including B_4Pr (2350 °C), where there are two eutectic reactions as the L↔(βPr)+B_5Pr_2 at 910 °C and L↔B_6Pr+(βB) at 2025 °C, and three peritectic reactions as the L+(βPr)↔(αPr) at 795 °C, L+B_4Pr↔B_5Pr_2 at 2000 ± 300 °C, and L+B_6Pr↔B_4Pr at 2350 °C.

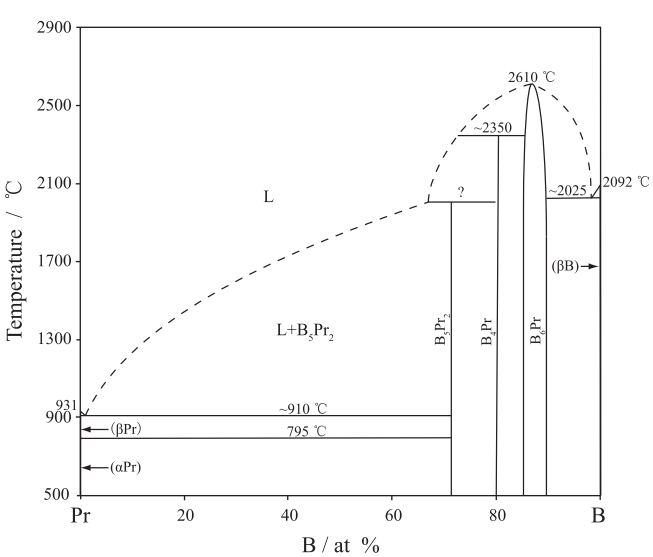

Fig. 2. The phase diagram of the B–Pr system reviewed by Moffatt [20].

In addition, the standard enthalpies of formation of the intermediate compounds of the B_4Pr and B_6Pr were measured by Meschel and Kleppa [19] using direct synthesis calorimetry (DSC) and solute–solvent drop calorimetry (SSC), respectively.

4. Optimized result and discussion

On the basis of lattices stabilities cited from Dinsdale [8], the optimization was carried out using the PARROT [24] module of Thermo-Calc software package developed by Sundman et al. [25]. This program is able to incorporate various kinds of experimental data in one operation. It works by minimizing the square sum of differences between experimental data and calculated values. In the optimization procedure, each of the selected data was given a certain weight. The weight was chosen and adjusted based upon the data uncertainties given in the original publications and upon the authors' judgment by examining all experimental data simultaneously.

In the present work, the parameters of intermediate compounds were optimized first because some experimental thermodynamic data and related phase boundaries are readily available, and then the parameters of the other terminal solutions and liquid phase were assessed. The evaluation was conducted one by one using the step-by-step optimization procedure. Reasonable agreement has been realized between the calculated results and experimental data.

4.1. B–Ce system

The calculated phase diagram of the B–Ce system compared with the experimental results is shown in Fig. 3. The calculated results are in excellent agreement with most of the experimental data [12] and [4,5]. Due to lack of thermochemical data for liquid, according to the relationship between the phase diagram and the free energy of formation of the compounds one can estimate several linear equations which contain the parameters to be determined. In addition, in the present work, solubility was neglected in the fcc, and bcc phases. This was realized by assigning a large positive interaction parameter, i.e.

$$^{0}L^{fcc}_{B,Ce} = {}^{0}L^{bcc}_{B,Ce} = 10000 \tag{9}$$

However, the calculated temperature (2000 °C) of the eutectoid reaction of the L↔B_6Ce+(βB) is lower than Stecher and Benesovsky's results (2200 °C) [12], and higher than Spear results (1900 °C) [4,5], but is in consistent with Seybolt's results [26]. In addition, the calculated liquidus between the liquid and B_4Ce phases slightly differ from the diagrams of Stecher and Benesovsky and Spear, and the similar phenomena happen between the liquid and B_6Ce phase. Future experiment is needed to confirm the calculated results. A complete set of the thermodynamic parameters describing the Gibbs free energy of each phase in this system is summarized in Table 2. All the invariant reactions in the B–Ce system compared with the experimental data are listed in Table 3.

The calculated standard enthalpies of formation of the intermediate compounds with reference states of γCe (bcc) and βB (rhombohedral) phases compared with the experimental data are shown in Fig. 4. Meschel and Kleppa measured the enthalpy of formation of the B_4Ce phase at 2380 °C by using the direct synthesis calorimetry (DSC) method [19]. There are no other thermochemical data available for the B_6Ce phase. It is seen that

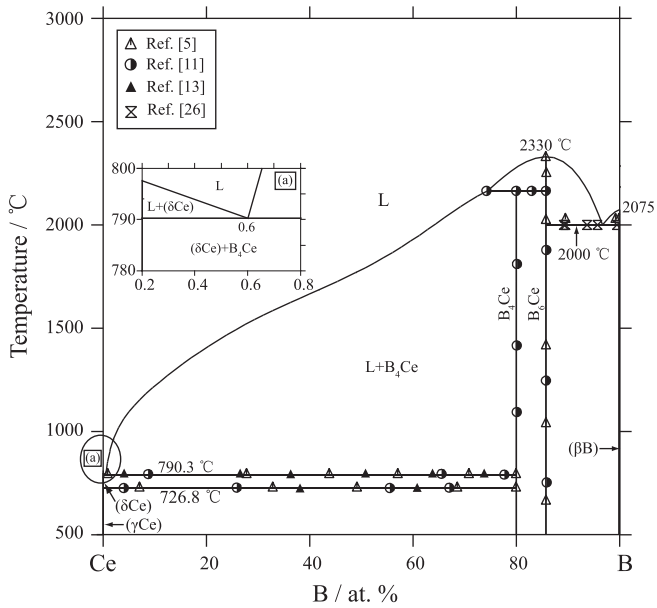

Fig. 3. Calculated phase diagram of the B–Ce system compared with the experimental data.

Table 2
Thermodynamic parameters in the B–Ce system assessed in the present work.

System	Parameters in each phase (J/mol)
B–Ce	Liquid phase, format (B, Ce)
	${}^{0}L^{Liq}_{B,Ce} = -71000-16T$
	${}^{1}L^{Liq}_{B,Ce} = -45000-23T$
	${}^{2}L^{Liq}_{B,Ce} -42000-23T$
	${}^{3}L^{Liq}_{B,Ce} = -29000$
	Fcc phase, format (B, Ce)
	${}^{0}L^{fcc}_{B,Ce} = 10000$
	Bcc phase, format (B, Ce)
	${}^{0}L^{Bcc}_{B,Ce} = 10000$
	B_4Ce phase, format $(B)_{0.8}(Ce)_{0.2}$
	$G^{B_4Ce}_{B:Ce} = -53765 + 2.7T + 0.8{}^{0}G^{rho}_{B} + 0.2{}^{0}G^{fcc}_{Ce}$
	B_6Ce phase, format $(B)_{0.857}(Ce)_{0.142}$
	$G^{B_6Ce}_{B:Ce} = -54320 + 4T + 0.857{}^{0}G^{rho}_{B} + 0.142{}^{0}G^{fcc}_{Ce}$

a reasonable agreement of the thermodynamic properties is obtained between the calculated results and experimental data.

4.2. B–Pr system

The calculated phase diagram of the B–Pr system together with the experimental results is shown in Fig. 5. There are no data available on the solubility of B in Pr–dhcp phase and Pr–bcc phase. Therefore, solid solubility of B in Pr–dhcp and of Pr in B-rhombohedral phase was assumed. In addition, solid solubility of B in Pr–bcc phase was realized by assigning a large positive interaction parameter, that is 10,000 for ${}^{0}L^{bcc}_{B,Pr}$. Since no experimental thermodynamic data for liquid were available, in the present work, the assessment was carried out based on the relationship between the phase equilibrium and the free energy of formation of the compounds. For the peritectic reaction of L $+B_4Pr \leftrightarrow B_5Pr_2$, Moffatt did not measure the temperature. But the melting temperature for B_5Pr_2 must be lower than that for B_4Pr, because the latter phase melts peritectically to give the B_6Pr plus a Pr-rich liquid [4,5]. Owing to the B_6Pr phase exhibits considerable solid solubility from 85.7% to 89.1% [20], the parameters were determined by phase equilibrium condition and the phase relationship between compounds and liquid. The calculated values are satisfactorily consistent with the measured data in Refs. [4,5]. Nevertheless, future experiment is needed to confirm the calculated results. A complete set of the thermodynamic parameters describing the Gibbs free energy of each phase in this system is summarized in Table 4. All the invariant reactions in the B–Pr system compared with the experimental data are listed in Table 5. Reasonable agreement has been realized between the calculated results and experimental data.

The calculated standard enthalpies of formation of the intermediate compounds with reference states of αPr (dhcp) and βB (rhombohedral) phases compared with the experimental data are shown in Fig. 6. Meschel and Kleppa measured the enthalpy of formation of the B_4Pr phase at 2350 °C and B_6Pr phase at 2610 °C by using direct synthesis calorimetry (DSC) method and solute–solvent drop calorimetry (SSC) [19]. But there are no other thermochemical data available reported for the B_5Pr_2 phase. It is seen that a reasonable agreement of the thermodynamic properties is obtained between the calculated results and experimental data.

Table 3
Comparison between the calculated and experimental results in the B–Ce system.

Invariant reaction	Reaction type	B (at%)			T (°C)	Reference
L↔(δCe)	Melting	–	0.0	–	798	[11]
		–	0.0	–	798.8	This work
L+(δCe)↔(γCe)	Peritectic	~0.1	~0.0	~0.0	726	[11]
		~0.1	~0.0	~0.0	726.8	This work
L↔(δCe)+B_4Ce	Eutectic	~1.0	~0.0	80.0	~790	[11]
		~1.0	~0.0	80.0	~790	[13]
		0.6	~0.0	80.0	790.3	This work
L+B_6Ce↔B_4Ce	Peritectic	74.0	85.7	80.0	2165	[11]
		74.4	85.7	80.0	2164.8	This work
L↔B_6Ce	Congruent	–	85.7	–	2330	[11]
		–	85.7	–	2330.3	This work
L↔B_6Ce+(βB)	Eutectic	~99.0	85.7	100.0	1990	[11]
		~96.0	85.7	100.0	2000	[26]
		96.7	85.7	100.0	2000	This work
L↔(βB)	Melting	–	100.0	–	2092	[18]
		–	100.0	–	2075	This work

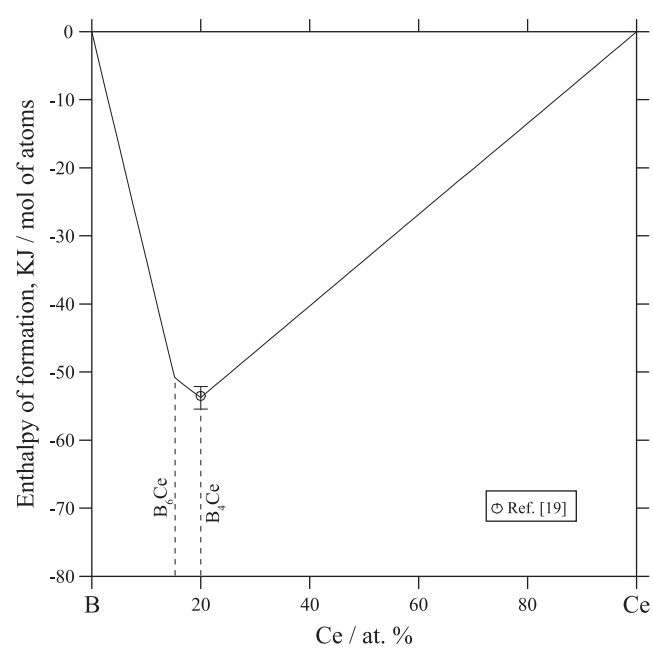

Fig. 4. Calculated standard enthalpies of formation with reference states of αCe (dhcp) and βB (rhombohedral) phases compared with the experimental data [19].

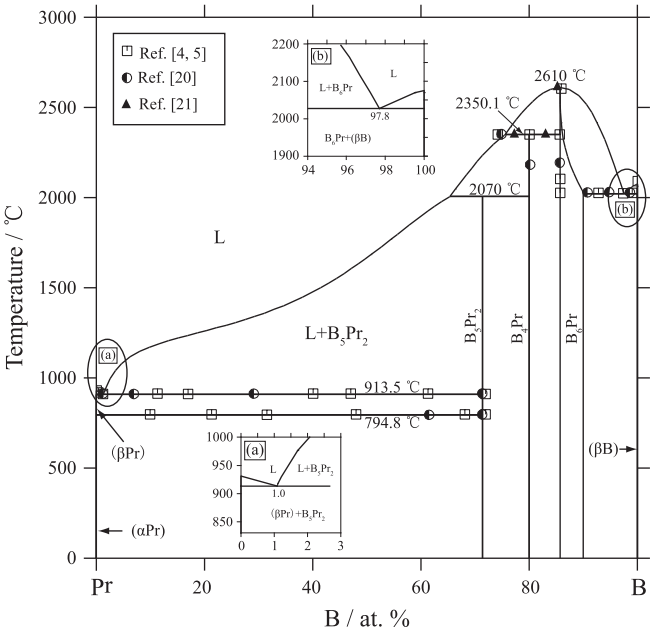

Fig. 5. Calculated phase diagram of the B–Pr system compared with the experimental data.

Table 5
Comparison between the calculated and experimental results in the B–Pr system.

Invariant reaction	Reaction type	*B* (at%)			*T* (°C)	Reference
L↔(βPr)	Melting		0.0		931	[20]
			0.0		931	This work
L+(βPr)↔(αPr)	Peritectic	~0.1	~0.0	~0.0	795	[4]
		~0.0	~0.0	~0.0	~795	[20]
		~0.0	~0.0	~0.0	794.8	This work
L↔(βPr)+B_5Pr_2	Eutectic	~1.0	~0.0	72.0	910	[4]
		~1.0	~0.0	71.4	~910	[20]
		1.0	~0.0	71.4	913.5	This work
L+B_4Pr↔B_5Pr_2	Peritectic	66.0	80.7	72.0	2000 ± 300	[4]
		66.0	80.0	71.4	–	[20]
		66.0	80.0	71.4	2070	This work
L+B_6Pr↔B_4Pr	Peritectic	74.0	85.7	80.0	2350	[4]
		75.0	85.7	80.0	~2350	[20]
		75.3	85.7	80.0	2350.1	This work
L↔B_6Pr	Congruent	–	85.7	–	2610	[4]
		–	85.7	–	2610	[20]
		–	85.7	–	2610	This work
L↔B_6Pr+(βB)	Eutectic	> 99.0	85.7	100.0	2025	[4]
		98.1	85.7	100.0	~2025	[20]
		97.8	85.7	100.0	2027.2	this work
L↔(βB)	Melting		100.0		2092	[4]
			100.0		2092	[20]
			100.0		2075	This work

Table 4
Thermodynamic parameters in the B–Pr system assessed in the present work.

System	Parameters in each phase (J/mol)
B–Pr	Liquid phase, format (B, Pr)
	${}^{0}L^{Liq}_{B,Pr} = -120000-21T$
	${}^{1}L^{Liq}_{B,Pr} = -110000-16T$
	${}^{2}L^{Liq}_{B,Pr} = -62000-5T$
	${}^{0}L^{Liq}_{B,Pr} = -24000$
	Bcc phase, format (B, Pr)
	${}^{0}L^{Bcc}_{B,Pr} = 10000$
	B_5Pr_2 phase, format $(B)_{0.714}(Pr)_{0.285}$
	$G^{B_5Pr_2}_{B:Pr} = -50686.625-6.5T + 0.714\,{}^{0}G^{rho}_{B} + 0.285\,{}^{0}G^{dhcp}_{Pr}$
	B_4Pr phase, format $(B)_{0.8}(Pr)_{0.2}$
	$G^{B_4Pr}_{B:Pr} = -53840-5T + 0.8\,{}^{0}G^{rho}_{B} + 0.2\,{}^{0}G^{dhcp}_{Pr}$
	B_6Pr phase, format $(B)_{0.8571}(B, Pr)_{0.1429}$
	${}^{0}G^{B_6Pr}_{B:B} = 8000 + {}^{0}G^{rho}_{B}$
	$G^{B_6Pr}_{B:Pr} = -54645-3T + 0.8571\,{}^{0}G^{rho}_{B} + 0.1429\,{}^{0}G^{dhcp}_{Pr}$
	${}^{0}G^{B_6Pr}_{B:B,Pr} = -14510 + 2T$

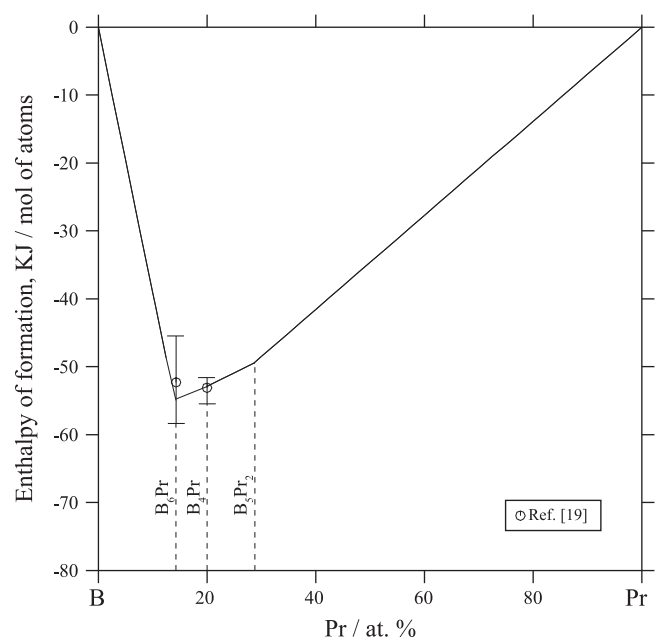

Fig. 6. Calculated standard enthalpies of formation with reference states of αPr (dhcp) and βB (rhombohedral) phases compared with the experimental data [19].

5. Conclusion

Thermodynamic optimization for the B–Re (Re: Ce, Pr) binary system is carried out by combining the thermodynamic models with the available experimental information in certain literature. A consistent set of optimized thermodynamic parameters has been derived for describing the Gibbs free energy of each solution phase and intermediate compound in the B–Re (Re: Ce, Pr) binary system, and good agreement between the calculated results and most of the experimental data is obtained.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Nos. 51031003 and 51171159), the Ministry of Education, PR China (No. 20120121130004), the National Key Basic Research Program of China (973 Program, No. 2012CB825700), and Fujian Provincial Department of Science and Technology (No. 2011HZ00004-1).

Appendix A. Supporting information

Supplementary information associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2013.03.001.

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